

NITYA MITTAL

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- American Express** | *Software Engineer - Associate DPM* July 2023 - July 2024
- **Saved ~\$70K** in Q1–Q2'24 through **33% data rationalization**, by designing & developing **Python** and **SQL** pipelines to resolve data quality issues and implementing optimization strategies for the **Hive** data layers used by the Anti-Money Laundering Transaction Monitoring team, Global Risk & Compliance Dept.
 - Led the implementation of onboarding pipelines for Kabbage acquisition, **migrating ~20K accounts** into Amex's banking ecosystem through scalable ingestion, validation, and reconciliation frameworks that ensured AML compliance.
 - Maintained and debugged production pipelines ingesting **millions of transactions** per 5-minute Amex feed, detecting file loss and duplication through transaction-volume anomaly monitoring in **Yellowbrick**; partnered with the AML Transaction Monitoring team via **JIRA** in an Agile environment to triage and resolve data-integrity issues.
- American Express** | *Data Engineering - Analyst Intern* May 2022 - July 2022
- Reduced development time by 80% for Model Performance Tracking of the Expected Credit Loss Forecasting Tool, by building **Python** scripts in **MLStudio** to consolidate all US and global portfolios at the Credit & Fraud Risk Dept.
 - Automated data sourcing across multiple formats (SQL, JSON) by designing object-oriented **Java** services with secure coding principles to pull and load data into **Hive**, following SDLC.
 - Streamlined onboarding for new data engineers by authoring technical design documentation in **Confluence** detailing BAU and ad-hoc workflows, sharing knowledge to support the team.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, TypeScript, JavaScript, PHP, SQL

Databases: PostgreSQL, Hive/HiveQL, Redis, Firestore

Cloud: AWS (EC2, S3, Lambda, ECS, CodePipeline); GCP (Vertex AI, Vertex Vector Search, Firestore)

DevOps & Infra: Git, Docker, Kubernetes, Apache Kafka, CI/CD, Jenkins, Linux

Web & Backend: REST APIs, Node.js, React, FastAPI, Streamlit, PHPUnit

ML & AI: scikit-learn, NumPy, pandas, PyTorch, TensorFlow, LangChain, transformers

EDUCATION

University of Michigan, Ann Arbor

Master of Science in Engineering in Computer Science

Aug. 2024 – May 2026

GPA: 4.0/4.0

Delhi Technological University, Delhi

Bachelor of Technology in Mathematics & Computer Science

July 2019 – May 2023

GPA: 3.81/4.0

Coursework: Data Structures, Algorithms, Operating Systems, Computer Networks, Database Management Systems, System Design, Object-Oriented Programming, Software Engineering.

PUBLICATION

“**University Recommendation System**” presented at **International Research Conference on Science Technology, Engineering and Management (IRCSTEM)**, 6–7 May 2023, Gurgaon, India.

PROJECT EXPERIENCE

Wikispeech (MediaWiki extension): Open Source Contribution

Jan 2026 – Jun 2026

- Built Wikimedia's article-to-audio download feature, enabling users to **save Wikipedia articles as speech**; delivered 4 **PHP** patches spanning schema design, **REST APIs**, JobQueue processing, and UI workflows, backed by 40+ **PHPUnit** tests.

CineRank: Real-Time Movie Recommendation System

Aug 2025 – Oct 2025

- Built a real-time recommendation engine serving movie suggestions in **<100ms latency** on the **MovieLens 32M** dataset. Designed a streaming ML pipeline using **Kafka**, **Redis**, **PostgreSQL**, & **FastAPI**, combining SVD collaborative filtering, cold-start fallback logic, and cached recommendation retrieval.

Needle: RAG-based Research Assistant

Aug 2025 – Nov 2025

- Built a research assistant for literature discovery & source-grounded Q&A over **1.7M+ arXiv papers** & user-uploaded PDFs, using retrieval-augmented generation to ground answers in retrieved sources. Engineered the pipeline on **Vertex AI** (Gemini 2.0 Flash + text-embedding-004), with vector search for semantic retrieval, **Firestore** for chunk metadata, and a **Streamlit** UI; integrated OpenCitations/Crossref to surface per-paper citation counts.

PrivNLGBench: Differential Privacy Benchmark

Jan 2025 – Jun 2025

- Built a benchmark for evaluating privacy-preserving NLP systems by transforming real-world datasets (Yelp, Enron, Clinton FOIA) into structured test sets with isolated PII and PII-free summaries. Developed the evaluation pipeline using **PyTorch**, **Opacus**, **DistilRoBERTa**, & **NetworkX**, achieving **0.85 accuracy** / **0.71 macro-F1** for PII detection and improving calibration from **0.272 to 0.083 ECE**.

ACHIEVEMENTS, EXTRACURRICULARS & LEADERSHIP

- **Awards:** McKinsey Next Generation Women Leaders 2023 & 2025 | **URAM (Micron) Scholarship** 2022 | **AWS ML Scholarship** 2021 | UCMAS International Distinction Grade 10
- **Leadership:** Judge, **MHacks 17 (MLH Official)** | **Vice Chairperson**, IEEE DTU CS | Student Advisory Board, Michigan Engineering | Networking Officer, GradSWE
- **Mentorship:** Mentored 300+ students at IEEE DTU, JA India & American Express, Joy of Coding (UMich)